

PLATINUM GAMES: THE PRESENT AND FUTURE

Tatsuya Minami is a 20-year veteran of the gaming

industry, having worked on the creative side of some of the biggest names around. But he's now running a company. Read on to find out how he does that in his own unique way
<http://dgit.in/13j60Js>

HOW DO COMPUTERS DECIPHER SPEECH?

The history of voice and speech recognition in

machines is a long and complicated one. Turns out that making computers understand our speech is one of the most difficult tasks, and this is how it's done
<http://dgit.in/173w1K6>

METACRITIC: THE BANE OF VIDEO GAMES?

Game review ratings are increasingly driving con-

sumers and game developers' decision to spend precious time and resources on a title. And this is hurting the video game industry, argues one gamer
<http://dgit.in/100jYXD>

SMARTWATCHES: THE NEXT BEST THING

The PC market is dying, and smartphones and tablets are

expected to have a bumper year in sales. But what's the next big thing? Read on to find out how tech giants are betting big on smartwatches and wearable tech
<http://dgit.in/113g3vF>

Shape-changing plastic could give touchscreens real physical clicky keyboards

- Graham Templeton



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Imagine if we could change the friction of a surface, make it smoother or bumpier with real physical deformations? More to the point, what if we could make a keyboard that actually popped out of the screen, and which could be depressed as we typed? Strategic Polymers claims it will bring a product to market next year that can do just that: pop up keys that actually click when clicked, and that do so with a ground-breaking millisecond response time.

Even *Star Trek* didn't posit such advances some 300 years in the future, with even the Federation's flagship using flat, chirpy touchscreens. The Strategic Polymers solution uses a new high-strain electromechanical material that can deform by "as much as 10%" and which responds quickly. Historically, we've had to choose one of those two virtues, getting either slow and meaningful responses, or quick and tiny ones. Here, Strategic Polymers claims we've got a technology that will allow true, clickable keys to pop right out of the surface of your smartphone's touchscreen, and to respond quickly and accurately. See an early video of the technology below.

The technology works via electrostriction, which is a property of every dielectric object in the world. By engineering this polymer to respond very specifically



to an applied electric field, however, the researchers claim to be able to create all sorts of useful deformations — like, for instance, a couple of dozen square keyboard keys. By adjusting the applied electric field in response to touch, the screen can seem to move in response to pressure, though in reality it's still only moving in response to the electric field, which is changing in response to touch. The result, if it all works as quickly and seamlessly as Strategic Polymers claims, would be a real physical keyboard on a fully functional touchscreen.

What is currently unclear, however, is how pre-programmed these physical features might need to be. It's of course exciting to imagine a touchscreen that could take on a new physical controller layout for every game or application, put a rough strip down the right of the screen for a nice tactile scroll bar, or volume slider you can feel slide, or perhaps just an embossed top logo. Current info doesn't make it clear if this will be possible, however, as the deformations might need to be built into the electro-

active polymer; a keyboard could work because a reliable key layout could be built right into the polymer. The ExtremeTech logo is, unfortunately, much more up in the air.

One very interesting feature is the material's ability to play sound. A speaker is just a highly controlled vibrating surface, really, one that creates patterns of tiny shockwaves that we interpret as sound. This haptic technology has a low enough response time that it can do this itself. This tech could turn just about any surface into a low-fi computer speaker.

Thin and transparent, the new tech is billed as yet another in the steadily growing list of screen layers. It's become a bit of design cliché to feature an exploded picture of a screen's many and innovative layers, from light scattering glass to touchscreen films to, now, haptic top coats. This layer will have to be on the surface of the screen, since it can't very well push up through a layer of Gorilla Glass, and so the claims about the material's durability will be absolutely key.

Virtually everyone who's used a touchscreen keyboard has wished for this technology at one time or another. It calls to mind all sorts of wild and questionably useful applications, like a raised ring you can physically stick your thumb into and drag down to scroll, or Google Maps with real topology. It's the sort of advance that could finally drag Apple out of the "But it's just a faster iPhone!" era, offering the biggest step forward in touchscreen functionality since multitouch.